

**Definition.**  $S \subseteq \mathbb{N}$  is called (register machine) **decidable** if its characteristic function  $\chi_S \in \mathbb{N} \rightarrow \mathbb{N}$  is a register machine computable function. Otherwise it is called **undecidable**.

So  $S$  is decidable iff there is a RM  $M$  with the property: for all  $x \in \mathbb{N}$ ,  $M$  started with  $R_0 = 0$ ,  $R_1 = x$  and all other registers zeroed eventually halts with  $R_0$  containing 1 or 0; and  $R_0 = 1$  on halting iff  $x \in S$ .

Basic strategy: to prove  $S \subseteq \mathbb{N}$  undecidable, try to show that decidability of  $S$  would imply decidability of the Halting Problem.